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 Studierichting: Informatica  
 Oefeningenreeks 1D nr. 6.2

$$(z^4 + 3z^2 - 1) \div (z^2 - iz)$$

|        |           |        |       |       |      |
|--------|-----------|--------|-------|-------|------|
| $z^4$  | $+3z^2$   | $-1$   | $z^2$ | $-iz$ |      |
| $-z^4$ | $+iz^3$   |        | $z^2$ | $+iz$ | $+2$ |
| $iz^3$ |           |        |       |       |      |
|        | $+3z^2$   | $-1$   |       |       |      |
|        | $-iz^3$   | $+z^2$ |       |       |      |
|        | $2z^2$    |        |       |       |      |
|        | $-1$      |        |       |       |      |
|        | $-2z^2$   | $+2iz$ |       |       |      |
|        | $2iz - 1$ |        |       |       |      |

**Quotiënt:**  $Q(z) = z^2 + iz + 2$

**Rest:**  $R(z) = 2iz - 1$

## References